

- OWEN NGO HANG CHAN, *A categorical characterisation of elimination of hyperimaginaries.*

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An early result in categorical model theory due to Makkai connects elimination of imaginaries for a theory to the exactness of the syntactic category of said theory. Namely, a complete theory T eliminates imaginaries if and only if the category $\text{def}(T)$ of definable sets and functions is exact. Building on Kamensky's work connecting projective limits and type-definable sets, we provide a categorical characterisation of elimination of hyperimaginaries in terms of the exactness of the pro-completion of $\text{def}(T)$.

[1] MOSHE KAMENSKY, *Ind- and pro- definable sets*, *Annals of Pure and Applied Logic*, vol. 147 (2007), no. 3, pp. 180–186.

[2] MICHAEL MAKKAÏ, GONZALO E. REYES, *First Order Categorical Logic: Model-Theoretical Methods in the Theory of Topoi and Related Categories*, Lecture Notes in Mathematics, vol. 611, Springer, Berlin, 1977.